

GLASGOW PRESTWICK, United Kingdom

WMO: 031350

Lat: 55.509N

Lon: 4.587W

Elev: 65

StdP: 14.66

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.4	26.8	20.4	15.3	27.8	22.9	17.3	29.9	40.0	44.2	35.7	45.8	1.9	110	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	73.4	62.9	69.9	61.6	67.7	60.4	64.5	71.2	62.9	68.2	61.3	65.9	8.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.2	83.7	68.0	60.6	79.3	66.2	59.0	74.8	64.2	29.6	71.4	28.4	68.5	27.4	65.9	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.5	25.6	22.6	DB	20.6	80.0	3.8	4.3	17.8	83.1	15.6	85.7	13.5	88.1	10.7	91.2	
			WB	19.6	67.6	3.9	2.1	16.7	69.1	14.4	70.3	12.2	71.5	9.4	73.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	40.8	41.1	43.2	47.1	51.7	56.4	59.5	59.2	55.8	50.8	45.0	41.1
	DBStd	8.46	5.52	5.22	4.75	4.73	4.90	4.04	3.65	3.67	4.07	4.87	5.68	6.82
	HDD50	1396	285	251	213	110	36	2	0	0	5	50	163	281
	HDD65	5725	749	669	675	538	413	259	177	185	277	442	599	742
	CDD50	1162	1	2	4	21	88	195	295	286	178	74	14	4
	CDD65	15	0	0	0	0	0	2	7	5	1	0	0	0
	CDH74	102	0	0	0	0	9	20	45	27	1	0	0	0
	CDH80	11	0	0	0	0	0	2	7	2	0	0	0	0
Wind	WSAvg	9.7	11.4	11.1	10.5	9.3	9.2	8.7	8.2	8.5	9.1	9.9	10.0	10.5
Precipitation	PrecAvg	37.4	3.6	2.6	2.9	2.0	2.1	2.4	2.7	3.3	4.1	4.3	4.1	3.8
	PrecMax	47.6	7.2	7.1	5.3	5.2	4.8	5.6	6.5	6.8	9.6	8.0	7.2	7.6
	PrecMin	27.8	0.5	0.2	0.7	0.2	0.3	0.8	0.6	0.7	0.8	1.4	1.0	0.9
	PrecStd	4.9	1.7	1.5	1.4	1.0	1.0	0.9	1.1	1.7	2.0	2.0	1.7	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	55.5	58.8	66.6	73.8	77.0	80.2	78.4	70.2	64.0	59.0	55.8
		MCWB	51.1	51.7	50.7	54.6	59.8	63.3	66.0	65.7	61.9	58.7	56.1	52.9
	2%	DB	51.9	52.1	53.9	60.7	68.1	70.1	73.4	71.6	67.6	61.0	56.9	53.9
		MCWB	49.4	49.0	49.0	52.3	57.8	60.7	63.8	63.4	60.8	57.3	54.0	51.0
	5%	DB	50.2	50.3	52.0	57.2	63.0	66.6	69.7	68.1	64.5	59.3	55.1	52.1
		MCWB	48.0	47.7	47.8	50.9	55.2	59.2	62.2	61.6	59.3	55.8	52.2	49.4
	10%	DB	48.5	48.5	50.2	55.0	60.5	64.3	66.4	66.0	62.5	57.6	53.4	50.2
		MCWB	46.7	46.3	46.9	49.5	53.8	57.5	60.2	60.1	57.9	54.2	50.6	48.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	52.6	52.7	55.8	61.8	64.2	67.4	67.0	64.2	59.8	56.2	54.0
		MCDB	54.2	55.3	56.7	63.1	71.6	71.9	75.5	75.0	69.2	62.6	58.4	56.1
	2%	WB	50.1	50.1	50.6	53.4	58.8	62.0	65.0	64.2	61.6	58.2	54.2	51.9
		MCDB	51.4	51.7	53.2	59.5	66.3	68.9	72.4	69.8	65.4	60.9	56.5	54.3
	5%	WB	48.4	48.3	49.0	51.5	56.3	59.9	62.9	62.6	59.9	56.6	52.5	50.3
		MCDB	49.7	50.1	51.3	56.2	62.1	66.0	68.3	67.0	63.5	59.3	54.8	52.1
	10%	WB	46.8	46.5	47.4	50.1	54.4	58.2	61.0	61.0	58.5	55.0	50.9	48.3
		MCDB	48.6	48.6	50.0	53.9	59.4	63.6	65.5	64.8	61.9	57.7	53.0	50.1
Mean Daily Temperature Range	5% DB	MDBR	8.7	10.1	11.3	13.2	14.4	12.9	12.6	12.4	11.9	10.8	9.9	9.7
		MCDBR	8.3	9.7	12.7	16.8	20.3	17.7	18.4	15.9	14.2	11.0	9.2	9.6
		MCWBR	8.1	8.5	9.3	10.8	12.1	10.5	10.7	9.7	9.4	8.8	8.4	9.1
	5% WB	MCDBR	8.3	8.6	10.9	14.8	17.9	16.2	15.6	13.9	11.9	10.0	9.2	9.6
		MCWBR	8.3	8.4	8.9	10.2	11.3	10.2	10.0	9.4	9.1	8.8	8.8	9.4
Clear-Sky Solar Irradiance	taub		0.338	0.348	0.373	0.390	0.393	0.388	0.389	0.373	0.365	0.348	0.328	0.323
	taud		2.202	2.296	2.271	2.260	2.301	2.352	2.356	2.411	2.429	2.402	2.289	2.183
	Ebn at Noon		174	226	249	263	269	272	269	266	252	225	180	149
	Edh at Noon		19	25	32	37	38	36	36	32	28	22	17	16
All-Sky Solar Radiation	RadAvg		156	374	734	1185	1517	1492	1429	1158	807	436	210	117
	RadStd		21	43	56	92	141	161	142	84	62	59	22	14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+1.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page